UNIVERSITY OF KALYANI

Department of Computer Science & Engineering

Comparative Analysis of Traditional Machine Learning and Transformer-Based Models for Sentiment Classification of Amazon Product Reviews

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DATA SCIENCE YEAR: 2024-26

DECLARATION

We hereby declare that the minor project entitled "**Comparative Analysis of Traditional Machine Learning and Transformer-Based Models for Sentiment Classification of Amazon Product Reviews"** submitted for the Master of Science(MSc Data Science) 3rd semester degree to University of Kalyani is a record of Bonafide work carried out by me under the supervision of (Prof).(Dr.) Debabrata Saddar, Department of Computer Science and Engineering, University of Kalyani, West Bengal.

We further declare that the work reported in this dissertation has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

Place: Kalyani, Nadia

Date:

Signature

Kaushik Bhattacharyya

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CERTIFICATE

This is to certify that the minor project entitled "**Comparative Analysis of Traditional Machine Learning and Transformer-Based Models for Sentiment Classification of Amazon Product Reviews"** submitted to the University of Kalyani in partial fulfilment of requirement for the degree of MSc (Data Science) 3rd semester in the Department of Computer Science and Engineering, is original work carried out by Kaushik Bhattacharyya(96/DTS/240010) and Sayan Mitra (96/DTS/240017).

The contents of this report have not been submitted and will not be submitted either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university. The dissertation fulfills the requirements and regulations of the University and, in my opinion, meets the necessary standards for submission.

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Date:

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University Of Kalyani

Signature of External Examiner

Acknowledgement

A project work is not an obligation done by one person towards a predetermined and very specific goal. Rather, it is the coming together of many elements, some direct and some indirect, towards the planning and implementation of a long-drawn effort. A project work requires hard work, perseverance, and dedication. And above all, a lot of support, as only teamwork can make good better and better best. So right at the onset, we would like to express our apologies to those whose names we have missed out to mention. Thanks to my parents who always believed and supported me in every way possible. We sincerely appreciate all the love, support, and trust, which made my work possible.

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Finally, yet importantly, my sincere thanks go to all who have directly and indirectly extended their valuable guidance and advice during the preparation of this report, which will give me the continuous flow of inspiration to complete this**.**

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Comparative Analysis of Traditional Machine Learning and Transformer-Based Models for Sentiment Classification of Amazon Product Reviews

Abstract

The rapid growth of e-commerce platforms has generated large volumes of user-generated product reviews, making automated sentiment classification essential for extracting actionable insights. This paper presents a comparative study of traditional machine learning and transformer-based approaches for sentiment classification on the Amazon Product Review dataset. Reviews were first labelled into positive, neutral, and negative classes based on star ratings. After text preprocessing, TF-IDF vectorization was applied to convert textual data into numerical features. A multiclass Logistic Regression model was implemented using different feature configurations, including review summaries, full review texts, and their combination. To enhance contextual understanding and model robustness, a pretrained Bidirectional Encoder Representations from Transformers (BERT) model was fine-tuned for sentiment classification. Experimental results indicate that TF-IDF-based Logistic Regression performs effectively on concise textual features, while its performance degrades for longer review texts. In contrast, the fine-tuned BERT model achieves higher accuracy and better generalization by capturing semantic and contextual dependencies. The study highlights the trade-offs between classical and transformer-based models for large-scale sentiment analysis.

1. Introduction

The rapid expansion of e-commerce platforms has resulted in a massive volume of user-generated product reviews, which play a crucial role in shaping consumer purchasing decisions and brand perception. Customers increasingly rely on textual reviews and ratings to assess product quality and overall satisfaction, making automated analysis of such data essential for both consumers and businesses. Extracting meaningful insights from large-scale review datasets enables organizations to better understand customer preferences and improve products and services accordingly.

Sentiment analysis, a core task in Natural Language Processing (NLP), aims to automatically classify opinions expressed in text as positive, neutral, or negative. However, accurate sentiment classification remains challenging due to informal language, contextual dependencies, and variations in review length. While traditional machine learning approaches use statistical representations such as TF-IDF, recent transformer-based models like BERT have achieved superior performance by capturing contextual and semantic relationships. This study investigates sentiment classification on the Amazon Product Review dataset by comparing classical machine learning techniques with fine-tuned transformer-based models to evaluate their effectiveness and limitations.

2. Background

Sentiment analysis has evolved from early rule-based and lexicon-driven methods to data-driven machine learning and deep learning approaches. Initial techniques relied on predefined sentiment lexicons and handcrafted rules, which lacked scalability and struggled with linguistic complexities such as negation, sarcasm, and domain-specific language.

The adoption of supervised machine learning algorithms, including Naïve Bayes, Support Vector Machines, and Logistic Regression, improved sentiment classification by learning patterns from labelled data. These models typically use bag-of-words or TF-IDF representations, which are effective for short text but fail to capture contextual and semantic relationships due to their independence assumption among words.

Word embedding methods such as Word2Vec and GloVe further advanced sentiment analysis by encoding semantic similarity in continuous vector spaces. However, these embeddings are context-independent and cannot distinguish word meanings across different contexts.

Transformer-based models, particularly BERT, addressed these limitations by generating context-aware embeddings using bidirectional self-attention mechanisms. Pretrained on large-scale corpora, BERT has achieved state-of-the-art performance across various NLP tasks, including sentiment classification. The Amazon Product Review dataset, with its scale and diversity, provides an ideal benchmark for evaluating and comparing traditional and transformer-based sentiment analysis approaches.

3. Problem Statement

Despite advancements in sentiment analysis, accurately classifying sentiment in large-scale e-commerce reviews remains challenging due to linguistic variability and contextual complexity in user-generated content. Amazon product reviews vary widely in length, writing style, and sentiment expression, often containing mixed, implicit, or neutral opinions that complicate automated classification.

The key challenges addressed in this research are:

- **Feature Representation Challenge:** Identifying the most effective textual features—review summary, full review text, or their combination—for accurate sentiment prediction.
- **Model Capability Gap:** Evaluating the limitations of traditional TF-IDF–based Logistic Regression in capturing contextual and semantic information compared to transformer-based models.
- **Performance vs. Complexity Trade-off:** Assessing whether the improved accuracy and robustness of fine-tuned BERT models justify their higher computational cost.
- **Model Selection Guidance:** Providing empirical insights to guide the selection of sentiment analysis models based on dataset characteristics and resource constraints.

This study aims to systematically compare traditional machine learning and transformer-based approaches for sentiment classification on the Amazon Review dataset, focusing on accuracy, generalization, and practical applicability.

4. Methodology

4.1 Data Labelling and Sentiment Categorization

The Amazon Product Review dataset contains numerical star ratings ranging from 1 to 5. To convert the problem into a supervised sentiment classification task, reviews were labeled into three sentiment categories based on their star ratings:

- **Negative sentiment: Ratings of 1 and 2**
- **Neutral sentiment: Rating of 3**
- **Positive sentiment: Ratings of 4 and 5**

This labelling strategy reflects common sentiment interpretation in e-commerce platforms and enables multiclass sentiment classification.

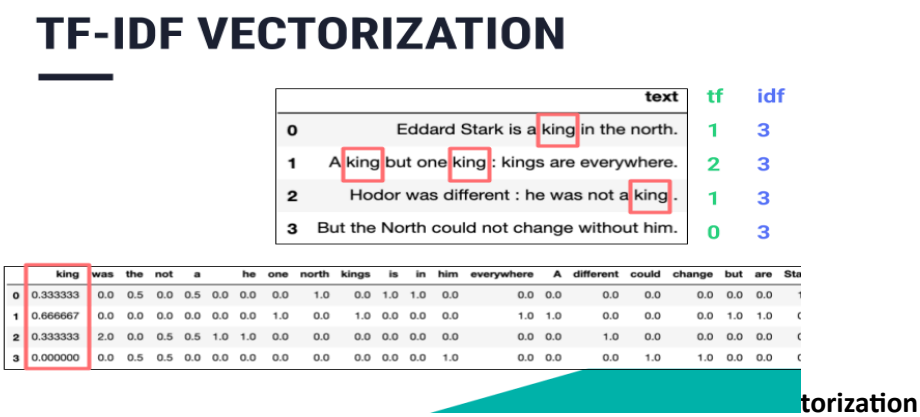
4.2 Text Preprocessing and Cleaning

Prior to feature extraction, a text cleaning function was applied to the review summaries and review texts. The preprocessing steps included:

- **Conversion of text to lowercase**
- **Removal of punctuation and special characters**
- **Elimination of numerical tokens**
- **Handling of missing values by replacing them with empty strings**

Text preprocessing was applied for traditional machine learning models to reduce noise and standardize textual input. For transformer-based models, minimal preprocessing was performed to preserve linguistic structure and contextual information.

4.3 Feature Extraction using TF-IDF Vec



	king	was	the	not	a		he	one	north	kings	is	in	him	everywhere	A	different	could	change	but	are	Sta
0	0.333333	0.0	0.5	0.0	0.5	0.0	0.0	0.0	1.0	0.0	1.0	1.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1
1	0.666667	0.0	0.0	0.0	0.0	0.0	0.0	1.0	0.0	1.0	0.0	0.0	0.0	1.0	1.0	0.0	0.0	0.0	1.0	1.0	0
2	0.333333	2.0	0.0	0.5	0.5	1.0	1.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1.0	0.0	0.0	0.0	0.0	0
3	0.000000	0.0	0.5	0.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1.0	0.0	0.0	0.0	1.0	1.0	0.0	0.0	0

torization

$$TF(t,d) = \frac{number\ of\ times\ t\ appears\ in\ d}{total\ number\ of\ terms\ in\ d}$$

$$IDF(t) = log\frac{N}{1+df}$$

$$TF-IDF(t,d) = TF(t,d) * IDF(t)$$

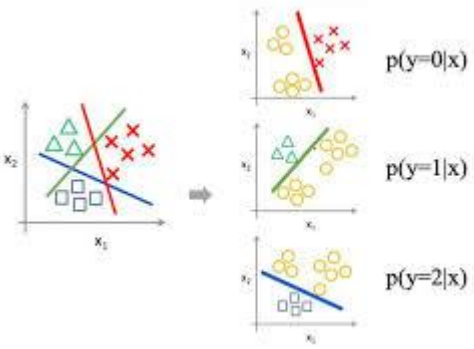
To convert textual data into numerical representations suitable for traditional machine learning models, Term Frequency–Inverse Document Frequency (TF-IDF) vectorization was employed. TF-IDF assigns weights to words based on their frequency within a document and their rarity across the corpus, emphasizing informative terms while down-weighting common words.

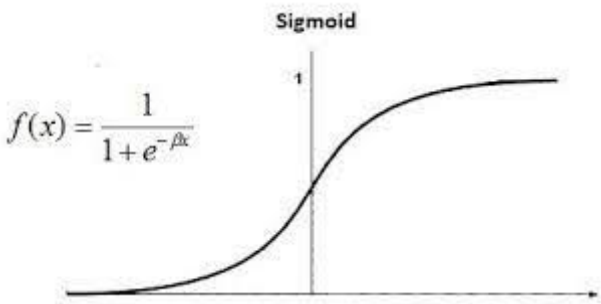
Separate feature sets were constructed using the following configurations:

- **Review summary only**
- **Full review text only**
- **Combined summary and review text**

This feature-wise experimentation enabled analysis of how different textual inputs influence classification performance.

4.4 Logistic Regression for Multiclass Sentiment Classification





Logistic Regression was used as the baseline machine learning model for multiclass sentiment classification. The model was trained using a one-vs-rest strategy to classify reviews into negative, neutral, and positive categories.

For each TF-IDF feature configuration, the Logistic Regression model was trained and evaluated independently. Performance metrics such as accuracy, precision, recall, and F1-score were computed to compare the effectiveness of different feature combinations.

4.5 Fine-Tuning Pretrained BERT for Sentiment Classification

To achieve improved accuracy and robustness, a pretrained Bidirectional Encoder Representations from Transformers (BERT) model was fine-tuned for the sentiment classification task. BERT's pretrained weights were initialized from a large-scale language corpus, and a task-specific classification head was added on top of the [CLS] token representation.

The fine-tuning process involved:

- **Tokenization of input text using the BERT tokenizer**
- **Preparation of PyTorch datasets for training and evaluation**
- **Training the classification head from scratch while fine-tuning BERT encoder weights**
- **Using a small learning rate to prevent catastrophic forgetting**

This approach enables the model to leverage contextual word representations and capture semantic dependencies within review text.

4.6 Model Evaluation and Comparison

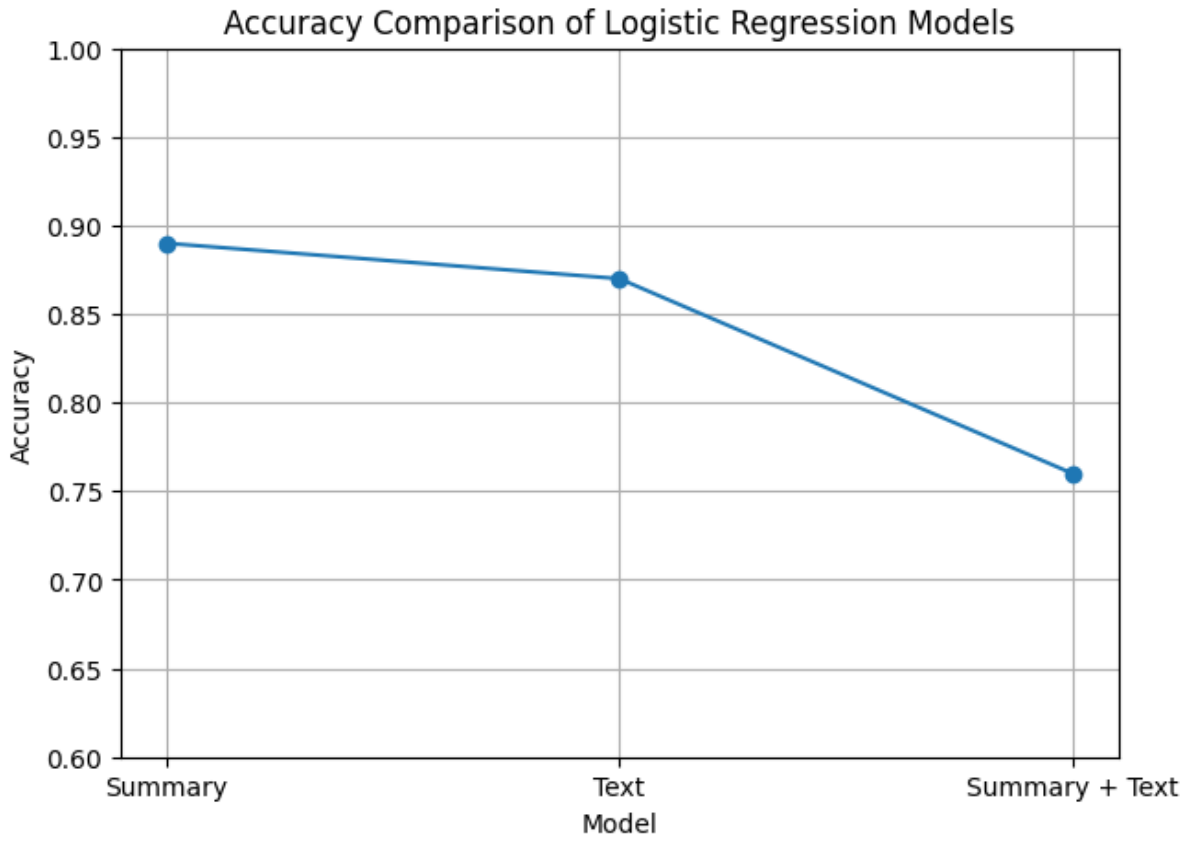
Both Logistic Regression and BERT-based models were evaluated using a held-out test set. Evaluation metrics included:

- **Accuracy**
- **Precision**
- **Recall**
- **F1-score**
- **Confusion matrix analysis**

Comparative analysis was performed to assess performance differences between traditional machine learning approaches and transformer-based models in terms of classification accuracy, generalization, and robustness.

5. Results and Discussion

This section presents a detailed analysis of the experimental results obtained from different sentiment classification models. The performance of Logistic Regression models trained on varying textual feature combinations is evaluated using accuracy, precision, recall, F1-score, and log loss. The results highlight the impact of feature selection on classification performance.



5.1 Logistic Regression Using Review Summary Only

The Logistic Regression model trained exclusively on the review summary achieved an overall accuracy of 89%, indicating strong predictive capability despite the limited textual input.

- **The model demonstrated excellent performance for positive sentiment, with a precision of 0.92, recall of 0.98, and F1-score of 0.95, showing that concise summaries often clearly express positive opinions.**
- **Negative sentiment was classified reasonably well, achieving an F1-score of 0.76.**
- **However, the model struggled with neutral sentiment, where recall dropped to 0.32, suggesting difficulty in distinguishing neutral opinions from positive or negative ones when only short summaries are used.**
- **The log loss of 0.3097 indicates well-calibrated probability estimates.**

```
... {'Predicted Sentiment': 'Positive', 'Confidence Scores': {'Negative': 0.0001, 'Neutral': 0.0018, 'Positive': 0.9981}}
```

Inference on unseen data further confirmed the model's reliability. For a strongly positive summary, the model predicted the correct sentiment with a high confidence score (99.81%), demonstrating robust probabilistic interpretation.

5.2 Logistic Regression Using Review Text Only

When trained solely on the full review text, the Logistic Regression model achieved an accuracy of 87%, slightly lower than the summary-based model.

- **The model captured richer contextual information, resulting in improved handling of negative sentiment (F1-score: 0.70).**
- **Positive sentiment remained dominant, with a strong F1-score of 0.93.**
- **Neutral sentiment performance remained weak (F1-score: 0.26), highlighting the inherent ambiguity of neutral expressions in long-form reviews.**
- **The log loss of 0.3563 suggests moderately confident probability predictions, though slightly less stable than the summary-only model.**

These results indicate that longer reviews introduce additional noise and mixed opinions, which can affect classification consistency.

5.3 Logistic Regression Using Combined Summary and Review Text

The model trained on the combined feature set (summary + review text) exhibited a significant performance drop, achieving only 76% accuracy.

- **While positive sentiment recall remained high (0.97), the model failed to effectively identify neutral and negative classes.**
- **Neutral sentiment recall dropped to nearly zero, and negative sentiment F1-score declined sharply to 0.05.**
- **The log loss increased to 0.7321, indicating poorly calibrated predictions and higher model uncertainty.**

This performance degradation suggests that simply concatenating summary and review text increases feature dimensionality and noise, leading to class imbalance dominance and reduced generalization.

5.4 Comparative Analysis and Key Observations

From the experimental results, several important observations can be drawn:

- **Summary-only features provided the best balance between accuracy, stability, and interpretability.**
- **Review text features captured richer context but introduced noise, slightly reducing performance.**
- **Feature combination without careful dimensional control adversely affected model learning.**

- **All Logistic Regression models performed best on positive sentiment, reflecting dataset imbalance and clearer linguistic cues.**
- **Neutral sentiment classification remains the most challenging, requiring deeper contextual modelling.**

Overall, the results demonstrate that traditional TF-IDF–based Logistic Regression can achieve strong performance with carefully selected features, while naive feature combination may degrade effectiveness.

5.5 Fine-Tuned BERT Model Performance

[22500/22500 1:14:06, Epoch 5/5]

Epoch	Training Loss	Validation Loss	Accuracy	F1
1	0.276000	0.392464	0.885250	0.872547
2	0.255800	0.448178	0.884500	0.869286
3	0.168000	0.448785	0.897250	0.893458
4	0.073300	0.565198	0.894250	0.892962
5	0.077800	0.682849	0.891750	0.891256

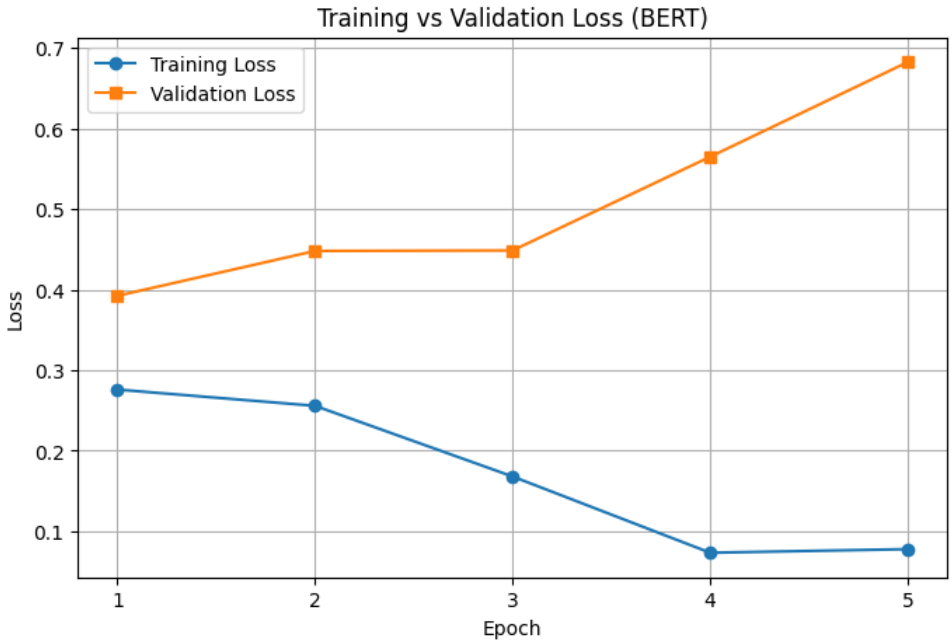
```
TrainOutput(global_step=22500, training_loss=0.19825708950890436, metrics={'train_runtime': 4605.2525, 'train_samples_per_second': 39.086,
'train_steps_per_second': 4.886, 'total_flos': 1.184010379776e+16, 'train_loss': 0.19825708950890436, 'epoch': 5.0})
```

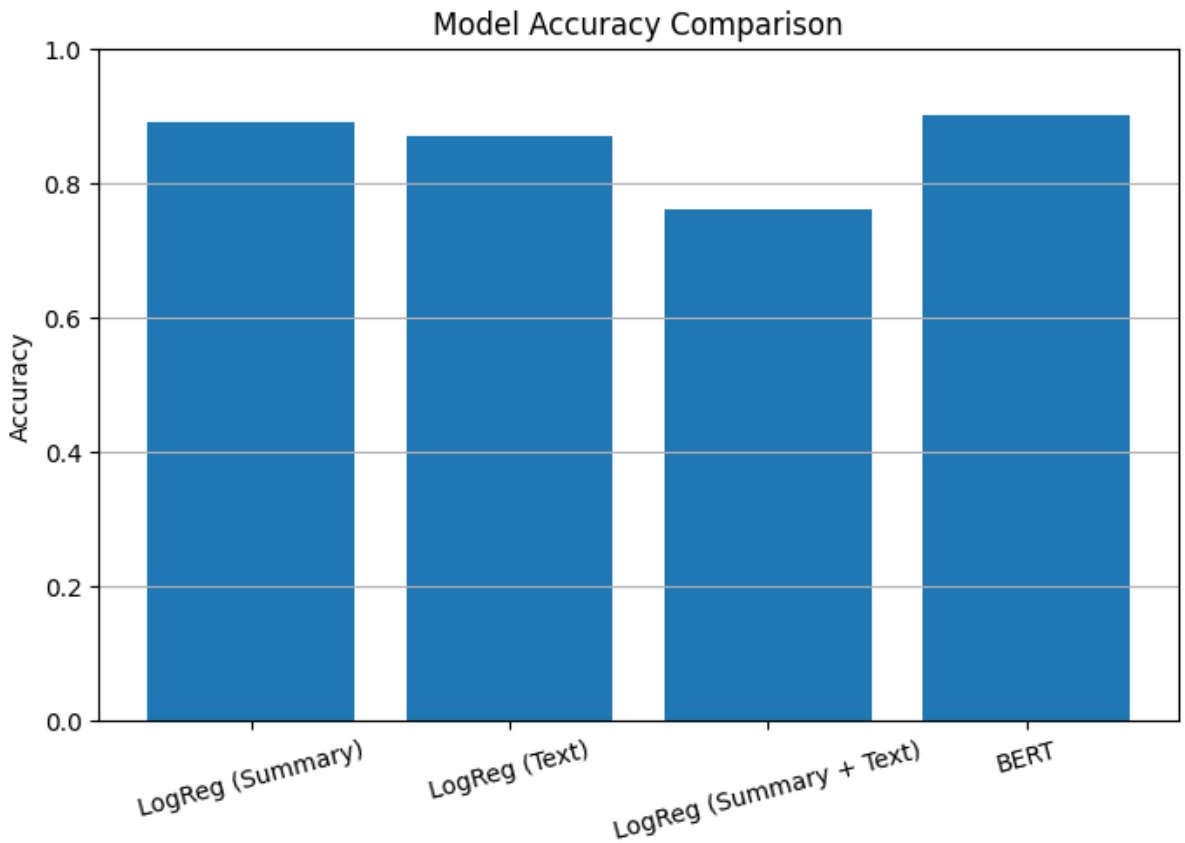
The fine-tuned BERT model achieved the highest overall accuracy of ~90%, outperforming all Logistic Regression variants.

- **Training loss consistently decreased, indicating effective learning.**
- **Validation loss increased after Epoch 3, suggesting mild overfitting, but accuracy and F1-score remained stable.**
- **Peak validation accuracy reached 89.7%, with strong F1-scores across epochs.**
- **BERT's contextual understanding enables better handling of neutral and mixed-sentiment reviews.**

The training–validation loss curve confirms that BERT captures deep semantic patterns but requires careful regularization and early stopping.

Key insight: Transformer-based models provide superior robustness and contextual understanding at the cost of higher computational complexity.


Training vs Validation Loss (BERT)
Training Loss
Validation Loss
Loss
Epoch
0.7
0.6
0.5
0.4
0.3
0.2
0.1
1
2
3
4
5


Model Accuracy Comparison
Accuracy
1.0
0.8
0.6
0.4
0.2
0.0
LogReg (Summary)
LogReg (Text)
LogReg (Summary + Text)
BERT

5.6 Comparative Performance Analysis

A visual comparison of model accuracies highlights the trade-off between performance and complexity:

Model	Accuracy
Logistic Regression (Summary)	89%
Logistic Regression (Text)	87%
Logistic Regression (Summary + Text)	76%
BERT	90%

- **Logistic Regression with summaries offers an excellent baseline.**
- **BERT provides incremental performance gains, particularly for nuanced sentiment.**

The combined TF-IDF model underperforms due to feature noise.

6. Conclusion:

- TF-IDF–based Logistic Regression provides a strong and efficient baseline for sentiment classification, achieving high accuracy when applied to concise review summaries.
- Performance of traditional models degrades on longer or combined text inputs due to limited contextual and semantic understanding.
- Fine-tuned BERT consistently outperforms traditional approaches by effectively capturing contextual dependencies in review text.
- Neutral sentiment remains the most challenging class across all models, indicating inherent ambiguity and class imbalance in real-world reviews.
- Traditional models are suitable for large-scale, low-resource deployments, while transformer-based models are preferable for accuracy-critical applications.

The comparative results offer actionable guidance for selecting sentiment analysis models based on performance needs and computational constraints.

7. Future Works:

- Fine-tune BERT on larger labelled data.
- Use class balancing techniques.
- Try advanced models(RoBERTa, DistilBERT).
- RNN & LSTM, Attention Mechanism ,Large Language Model.
- Aspect-based sentiment analysis.

8. References:

- **Fine -Tuning BERT for Robust Sentiment Classification of IMDB Reviews [Orestis Papadimitriou] [Khalil AL-Hussaeni] [Manolis Maragoudakis]**
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- **Fine-tune BERT based on Machine Learning Models for Sentiment Analysis [Nadia Smairi] [Houda Abadlia]**
  https://share.google/FqULgEX7t2XNGcQC2